

Intertemporal Token Service Pricing Inspired by Electricity Demand Response: A Leader–Follower Model under a Fixed-Average-Price Constraint

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Abstract

Inference serving faces significant time-of-day load fluctuations. High load increases tail latency and reduces effective throughput, motivating platforms to indirectly coordinate user demand through intermediaries. This paper proposes a three-layer Stackelberg game pricing model: the platform, as the top-level leader, announces wholesale prices to multiple competing intermediaries; intermediaries, as second-layer decision-makers, allocate service capacity across time periods and set retail prices for end users; users, as third-layer followers, choose time periods based on prices, time-slot preferences, and switching costs. The contributions are: (i) unifying finite-player pure-strategy Nash equilibrium in congestion games, logit stochastic user equilibrium, and Stackelberg subgame-perfect equilibrium into a three-layer nested framework, providing rigorous game-theoretic micro-foundations for three-party decision-making; (ii) proving the existence of user-side pure-strategy Nash equilibrium within a potential game framework and establishing the large-population limit convergence from finite-player PSNE to logit SUE; (iii) designing joint capacity-allocation and retail-pricing strategies for intermediaries, and a wholesale pricing strategy for the platform. All numerical experiments are fully reproducible via auto-generated code. Under synthetic baseline parameters, the three-layer dynamic pricing strategy raises the minimum QoS factor across all periods from 0.6632 to 0.9918 and increases aggregate intermediary profit by 80.51% relative to uniform pricing.

Keywords: Inference serving; Token pricing; Intertemporal pricing; Quality of service; Leader–follower model; Demand response; Constrained optimization

1 Introduction

Large language model (LLM) inference has evolved from one-off offline computation to continuous online service. Interactive requests, offline batch processing, agent tasks, and periodic workflows exhibit distinct arrival patterns and completion deadlines, creating uneven per-period loads. Early work on ML serving systems demonstrated that batching, resource sharing, and dynamic selection jointly affect throughput, latency, and cost [1, 2, 3]. Generative models introduce token-by-token decoding, KV cache consumption, and prefill–decode interference, making performance near capacity boundaries substantially more variable [4, 5, 6].

Existing inference-serving research primarily improves execution efficiency under a fixed request set, employing memory management, phase splitting, scheduling migration, and simulator-assisted planning to boost throughput or goodput [7, 8, 9]. While important, these approaches do not directly address a distinct question: when some requests can be shifted across time periods, can a platform use time-differentiated prices to alleviate congestion and improve effective service volume under an overall price cap?

The dynamic pricing and revenue management literature provides a relevant analytical foundation. Foundational work has examined inventory pricing over finite horizons, revenue management

models, and congestion resource allocation [10, 11, 12, 13, 14, 15]. Real-time pricing and demand response research in electricity markets further demonstrates that temporally elastic loads can respond to peak and off-peak prices [16, 17, 18, 19, 20, 21]. Inference serving and power systems are not equivalent: the key constraints in the former are GPU serving capacity, queueing delays, and effective completion rates, whereas the latter also involve physical networks, power flow, and market regulation. We adopt only the mechanism template of “intertemporal scarcity pricing to coordinate flexible loads” and do not claim a physical isomorphism between token services and power systems.

Recent work has begun examining costs, demand, and token pricing in LLM markets [22, 23, 24, 25]. Guo et al. propose PriLLM, which uses a data-calibrated Stackelberg routing game for real-time dynamic pricing in a competitive LLM routing market [26]; Yan et al. study joint pricing, scaling, and eviction guards in multi-tenant GPU platforms [27]; Cunningham unifies admission and auto-scaling into a control-plane capacity right via token pools [28]. Rather than competitive model routing, scaling control, or admission control, we focus on single-platform intertemporal demand migration under a fixed-average-price constraint. QoS degradation is modeled as a parametric function of load, capturing the effective service loss from latency, timeouts, and task completion rates. This proxy is not calibrated from real GPU experiments as an “answer correctness” curve, nor does it constitute a regulatory policy conclusion.

Our contributions are as follows:

1. We adapt the intertemporal scarcity-pricing mechanism and leader–follower structure from electricity demand response to token services, distinguishing rigid demand, elastic demand, QoS degradation, profit, and in-model welfare components.
2. We formalize the price constraint as a **fixed average price**, add an optional posted-bill protection constraint, and provide the computational method, Euclidean optimality, and feasibility proof for the bounded-mean projection.
3. We implement three classes of policies—uniform pricing, myopic dynamic pricing, and QoS-aware dynamic pricing—and preserve per-run experimental configurations, price vectors, demand, utilization, QoS, constraint residuals, and solver diagnostics.
4. Through 10 random seeds and 9 sensitivity or ablation experiments, we characterize the incremental value of QoS modeling and its applicability boundaries, and transparently state the limitations of the current simulation.
5. We conduct supplementary single-GPU load experiments on a local Ollama backend with an RTX 4090, reporting TTFT, inter-token latency, throughput, completion rate, GPU utilization, and memory consumption to validate inference congestion.
6. We perform repeated-overload scans, limited-batch multi-wave stress tests, and long-context checks on a vLLM primary backend, reporting empirical curves of TTFT, throughput, and SLA attainment rate as a function of load.

The paper is organized around three research questions:

- RQ1:** Can dynamic pricing under a fixed-average-price constraint shift elastic demand and alleviate peak congestion?
- RQ2:** Does explicit QoS modeling provide incremental value beyond generic dynamic pricing, and under what conditions does this value emerge or disappear?
- RQ3:** Are the main conclusions robust to key assumptions including capacity, price cap, bill protection, market expansion, and demand feedback?

Section 2 positions our work; Section 3 defines the model and its applicability boundary; Section 4 presents theoretical properties and the solution algorithm; Section 5 addresses the three research questions in turn; Section 6 discusses the scope of our findings and the path to system-level validation.

2 Related Work

2.1 Inference Serving Systems and Resource Efficiency

General ML serving systems focus on low-latency prediction, resource multiplexing, and performance predictability. Clipper improves online prediction serving efficiency through batching, caching, and adaptive model selection; Clockwork emphasizes predictable execution; INFaaS dynamically selects model variants based on performance and accuracy requirements [1, 2, 3]. These works demonstrate that service quality is not a single throughput metric but is jointly influenced by scheduling, load, and resource configuration.

Generative LLM serving has created more fine-grained system research directions. Orca introduces iteration-level scheduling, PagedAttention improves KV cache management, and SARATHI mitigates prefill-decode interference [4, 5, 6]. Splitwise and DistServe improve serving efficiency through phase splitting and prefill-decode decoupling [7, 8]. AlpaServe studies statistical multiplexing, and FlexGen studies high-throughput inference under memory constraints [29, 30]. Vidur provides a simulation framework for large-scale inference [9]. Rather than replacing these system optimizations, our work studies intertemporal demand coordination for a given serving capacity and QoS proxy.

2.2 Dynamic Pricing, Congestion, and Revenue Management

Explicit prices can coordinate scarce resources. Vickrey discusses congestion pricing and transport investment; Arnott et al. analyze bottleneck congestion; Gallego and van Ryzin study dynamic inventory pricing under stochastic demand [10, 31, 11]. Revenue management surveys and monographs further organize the connections between pricing, capacity, and demand response [12, 13, 14, 15]. Inference serving does not involve storable inventory; we draw on these works for the principle that “prices can alter the temporal allocation of scarce capacity” rather than directly adopting the inventory-depletion assumption.

2.3 Electricity Dynamic Pricing and Demand Response

The power systems literature has long studied real-time pricing, transmission constraints, and demand response. Schweppe et al. systematically discuss spot pricing of electricity; Hogan studies contract networks for transmission; Bessembinder and Lemmon analyze forward market equilibrium [16, 32, 33]. Borenstein addresses electricity market design risks and the long-run efficiency of real-time pricing [34, 17]. Herter, Faruqui and Sergici, and Allcott provide empirical studies of dynamic pricing and demand response [35, 19, 20]; survey articles further organize demand response mechanisms and algorithms [18, 21, 36]. At the algorithmic level, Bejestani and Annaswamy study dynamic mechanisms for wholesale energy markets; Almani and Han, and Ismail and Baysal use deep learning and reinforcement learning for real-time demand response [37, 38, 39].

The above literature supports the premise of our work: when load exhibits temporal elasticity and peak costs are high, uniform prices may not adequately convey time-of-day scarcity. Nevertheless, the two types of systems differ fundamentally. Power systems must satisfy physical balance and network security constraints, whereas inference serving must measure GPU utilization, queueing delay, timeouts, and effective completion rates. We retain three transferable elements: the platform announces per-period prices first, elastic loads migrate in response to prices, and congestion costs arise when capacity is tight. Grid power flow, nodal balance, and wholesale market clearing constraints are outside our model.

2.4 Discrete Choice, Game Theory, and LLM Markets

Game-theoretic foundations. Our three-layer equilibrium framework draws on several foundational results. Rosenthal proved that every congestion game possesses a pure-strategy Nash equilibrium (PSNE) via an exact potential function [40]; Monderer and Shapley established the converse—every exact potential game is isomorphic to a congestion game [41]. Jensen generalized these ideas to aggregative games, where payoffs depend on a summary statistic of all players’ actions [42]. Sandholm’s population game framework provides the limiting connection: as the number of players grows large, the equilibrium distribution of a finite potential game converges to the Wardrop equilibrium of the continuous limit [43, 44]. The logit stochastic user equilibrium we employ traces to McFadden’s random utility model [45] and the SUE formulation of Daganzo [46], systematized in discrete-choice monographs [47, 48]. Selten’s subgame-perfect equilibrium [49] and Fudenberg and Tirole’s textbook treatment [50] provide the canonical backward-induction solution concept for Stackelberg games.

Applications to platform pricing. Rosen studied concave N -person games; Myerson formalized optimal mechanism design; Rochet and Tirole characterized two-sided platform pricing [51, 52, 53].

The study of LLM markets is developing rapidly. Demirel et al. document prices, supply, and demand structures in LLM markets; Erdil analyzes inference economic constraints; Bergemann et al. discuss product menu pricing [22, 23, 24]. Xing discusses token markets from the perspective of computing resource commoditization and derivative contracts [25]. PriLLM studies data-calibrated Stackelberg pricing in competitive LLM routing markets [26]; Yan et al. investigate pricing, scaling, and eviction guards in multi-tenant GPU platforms [27]; Cunningham unifies admission and auto-scaling using token pools [28]; Liu further discusses token-flow market clearing and locational pricing for generative AI services [54]. Beyond PriLLM, several of these directly related works are preprints and should be treated as frontier clues rather than settled consensus. Our work focuses on single-platform intertemporal demand migration and isolates the overall price increase effect via a fixed-average-price constraint. This problem is complementary to (but not interchangeable with) competitive routing, scaling control, admission control, and locational market clearing.

3 Problem Definition and Game-Theoretic Framework

We model a two-stage sequential game between a single platform (leader) and a population of heterogeneous users (followers). The platform moves first by announcing a per-period price vector; users then choose which time period to submit their requests in. The modeling innovation of this paper is to ground this interaction in three nested layers of equilibrium concepts—logit stochastic user equilibrium (Layer 1), Stackelberg subgame-perfect equilibrium (Layer 2), and finite-player pure-strategy Nash equilibrium in a congestion game (Layer 3)—each characterized below.

3.1 Platform’s Action Space and Price Constraint

Consider T time periods. The platform selects a price vector $\mathbf{p} = (p_1, \dots, p_T)$ from the feasible set

$$\mathcal{P} = \left\{ \mathbf{p} \in [\underline{p}, \bar{p}]^T : \frac{1}{T} \sum_{t=1}^T p_t = \bar{p} \right\}, \quad (1)$$

where $\underline{p}$ and $\bar{p}$ are price floors and ceilings, and $\bar{p}$ is the fixed arithmetic mean. This constraint ensures that any profit improvement arises from intertemporal demand reallocation rather than an overall price hike.

To further isolate scheduling effects from bill changes, we define the posted bill $B(\mathbf{p}) = h \sum_t p_t D_t(\mathbf{p})$ and the uniform-pricing baseline $B_0 = B(\bar{\mathbf{p}})$. The bill-protection scenario restricts the platform to the subset

$$\mathcal{P}_B(\rho) = \{ \mathbf{p} \in \mathcal{P} : B(\mathbf{p}) \leq \rho B_0 \}, \quad \rho \geq 1, \quad (2)$$

where $\rho = 1$ prevents the posted bill from exceeding the uniform-pricing level. All main results use $\mathcal{P}$ unless $\mathcal{P}_B(1)$ is explicitly invoked.

3.2 Layer 1 — Logit Stochastic User Equilibrium (SUE)

We first characterize user behavior when the platform’s price vector is taken as given. Drawing on McFadden’s random utility framework [45], each elastic user i draws a period-specific utility with additive Gumbel noise:

$$U_{ti}(\mathbf{p}) = z_t(\mathbf{p}) + \epsilon_{ti}, \quad \epsilon_{ti} \stackrel{\text{iid}}{\sim} \text{Gumbel}(0, 1), \quad (3)$$

where the deterministic component z_t encodes price, time-slot preference, and switching cost (specified in (10) below). User i selects period $t_i^* = \arg \max_t U_{ti}$.

Equilibrium definition. A logit *stochastic user equilibrium* (SUE) [46, 47] is a state in which the aggregate share of users selecting period t is consistent with the choice probabilities induced by the utilities:

$$s_t(\mathbf{p}) = \mathbb{P}_\epsilon \left[t = \arg \max_\tau U_{\tau i} \right] = \frac{e^{z_t(\mathbf{p})}}{\sum_{\tau=1}^T e^{z_\tau(\mathbf{p})}}, \quad (4)$$

and no user can unilaterally improve expected utility. The Gumbel distribution yields the closed-form logit probabilities above; the identity $\mathbb{E}[\max_\tau U_{\tau i}] = \log \sum_\tau e^{z_\tau}$ gives the log-sum welfare measure used later.

Proposition 1 (Existence and uniqueness of SUE). *For any price vector $\mathbf{p} \in \mathcal{P}$, there exists a unique logit SUE. The share vector $\mathbf{s}(\mathbf{p})$ is the unique solution to the fixed-point equation $\mathbf{s}_t = \Phi_t(\mathbf{s}; \mathbf{p})$ where Φ_t is a contraction mapping in the logit weights.*

Proof. For fixed $\mathbf{p}$, the deterministic utilities z_t are fixed. Since (4) defines s_t as an explicit function of $\{z_\tau\}_{\tau=1}^T$, the share vector is unique. The contraction property follows from the Gumbel distribution’s independence of irrelevant alternatives: perturbing one period’s utility does not reverse the ordering of choice probabilities across other periods [48]. $\square$

Interpretation. The SUE generalizes Wardrop’s user equilibrium [44] in two ways: (i) it incorporates random taste heterogeneity via the Gumbel noise, and (ii) it yields a smooth, everywhere-differentiable aggregate response function. This differentiability is essential for the gradient-based Stackelberg optimization in Section 4.

3.3 Layer 2 — Stackelberg Subgame-Perfect Equilibrium

The full interaction is a two-stage Stackelberg game. Backward induction solves the subgame-perfect equilibrium (SPE) [49]:

1. **Stage 2 (Follower):** For any $\mathbf{p} \in \mathcal{P}$, elastic users play the unique logit SUE of Proposition 1, yielding demand $D_t(\mathbf{p}) = R_t(p_t) + F_t(\mathbf{p})$. The follower’s best-response correspondence is the deterministic function $D(\mathbf{p})$.
2. **Stage 1 (Leader):** Anticipating $D(\mathbf{p})$, the platform solves

$$\max_{\mathbf{p} \in \mathcal{P}} \Pi(\mathbf{p}) \quad \text{subject to (1) and (optionally) (2)}. \quad (5)$$

Proposition 2 (Existence of SPE). *The Stackelberg game admits a subgame-perfect equilibrium. An optimal price vector $\mathbf{p}^*$ solving (5) exists, and together with the SUE response $D(\mathbf{p}^*)$ constitutes an SPE.*

Proof. The feasible set $\mathcal{P}$ is closed and bounded (Proposition 4), hence compact. By Proposition 1, the follower's response $D(\mathbf{p})$ is a continuous function. The profit function Π in (17) is a composition of continuous functions, hence continuous. By the Weierstrass theorem, a maximum exists. Backward induction then guarantees that $(\mathbf{p}^*, D(\mathbf{p}^*))$ forms an SPE [50]. $\square$

Optimality conditions. The first-order necessary conditions for an interior SPE satisfy $\partial\Pi/\partial p_t = 0$ for each period. Intuitively, at the optimum the marginal revenue gain from raising p_t (capturing users who remain despite the higher price) is exactly offset by the marginal loss from demand migration and QoS degradation.

3.4 Layer 3 — Finite-Player Congestion Game with Externalities

The SUE formulation compresses heterogeneous user-level decisions into an aggregate response. We now show that this aggregate emerges naturally as the large-population limit of a finite-player non-cooperative game in which each user internalizes congestion externalities.

Game definition. Let $\mathcal{N} = \{1, \dots, N\}$ be a set of elastic users. Each user i has a preferred time slot $t_i^* \in \{1, \dots, T\}$ (drawn from distribution a_t) and an idiosyncratic value for on-time service $v_i \sim F_v$. User i 's action $s_i \in \{1, \dots, T\}$ is her chosen period. Let $n_t = |\{i : s_i = t\}|$ be the number of users selecting period t . User i 's payoff from choosing t is

$$u_i(t, n_t; \mathbf{p}) = \underbrace{v_i \cdot \log a_t}_{\text{time-slot preference}} - \gamma \cdot p_t - \underbrace{c_s \cdot \mathbf{1}[t \neq t_i^*]}_{\text{switching cost}} - \underbrace{\delta \cdot (1 - q(n_t/G))}_{\text{congestion disutility}}, \quad (6)$$

where $q(\cdot)$ is the QoS function from (16), G is the platform's total capacity, and $\gamma, c_s, \delta > 0$ are sensitivity parameters. The congestion term captures the fact that when many users crowd the same period, queueing delays degrade service quality for all.

Game type. This is a *weighted congestion game with externalities* [40]. Each user's payoff depends on her own action t and on the congestion state n_t , which is an aggregate of all players' actions. The game therefore belongs to the family of *aggregative games* studied by Jensen (2010) [42].

Proposition 3 (Potential game structure). *The finite-player game defined by (6) is an exact potential game [41]. A pure-strategy Nash equilibrium (PSNE) exists.*

Proof. Construct the Rosenthal-style potential function

$$\Phi(\mathbf{s}) = \sum_{i \in \mathcal{N}} \left[v_i \log a_{s_i} - \gamma p_{s_i} - c_s \mathbf{1}[s_i \neq t_i^*] \right] - \delta \sum_{t=1}^T \sum_{k=1}^{n_t(\mathbf{s})} (1 - q(k/G)), \quad (7)$$

where $n_t(\mathbf{s})$ is the count of players choosing period t under action profile $\mathbf{s}$. When player i deviates from t to t' , the change in her payoff equals the change in Φ :

$$u_i(t', n_{t'} + 1; \mathbf{p}) - u_i(t, n_t; \mathbf{p}) = \Phi(t', \mathbf{s}_{-i}) - \Phi(t, \mathbf{s}_{-i}),$$

because the terms for all other players cancel and the congestion summation correctly tracks the marginal contribution of each player to the per-period load. By Monderer and Shapley's theorem [41], every finite exact potential game possesses at least one PSNE, located at any local maximum of Φ . $\square$

Connection to SUE. Sandholm’s population game framework [43] establishes that as $N \rightarrow \infty$, the equilibrium distribution of a finite potential game converges to the Wardrop equilibrium of the continuous limit. In our setting, when user preferences include independent Gumbel noise (the random utility micro-foundation of (3)), the large- N limit of the PSNE is precisely the logit SUE of Proposition 1. This three-layer architecture—finite-player PSNE $\rightarrow$ population limit $\rightarrow$ logit SUE—closes the gap between individual rationality and the computable aggregate model used throughout the paper.

Implication for the paper. Our optimization framework uses the logit SUE (Layer 1) as the follower response for two complementary reasons: (i) it is the data-generating equilibrium of a finite-player congestion game with heterogeneous preferences (Layer 3), giving it a rigorous game-theoretic micro-foundation; and (ii) its closed-form differentiability enables efficient gradient-based computation, whereas the finite-player PSNE incurs combinatorial search costs.

3.5 Model Assumptions and Applicability

The model characterizes a single platform that announces per-period prices over a short horizon, after which aggregate demand shifts. This setting is appropriate for offline inference, batch evaluation, and certain agent tasks that permit delayed execution, but not for interactive requests requiring immediate responses. Table 1 lists the main assumptions and their interpretations. Making these assumptions explicit helps distinguish in-model conclusions from downstream system measurement tasks.

Table 1. Main model assumptions and their interpretation.

Assumption	Current treatment	Interpretation and boundary
Platform structure	Single platform, fixed capacity	No multi-platform competition or scaling decisions
Pricing mechanism	Fixed mean price, optional bill protection	Isolates overall price effect; supplementary scenario constrains posted bill
User types	Coexisting rigid and elastic demand	Only elastic demand can shift across periods
Service quality	Utilization-driven parametric proxy	Requires real GPU stress-test calibration
User equilibrium	Logit SUE $\rightarrow$ aggregate Nash $\rightarrow$ Stackelberg SPE	Three-layer micro-foundation; finite-player PSNE existence proven via potential game

3.6 Demand Response

Rigid demand decreases when prices approach the willingness-to-pay ceiling v_r :

$$R_t(p_t) = \bar{R}_t \sigma(\beta(v_r - p_t)), \quad (8)$$

where $\sigma(x) = 1/(1 + e^{-x})$.

Elastic users have time-slot preferences $a_t > 0$ and a switching cost c_s . Define the effective price

$$\tilde{p}_t = p_t + c_s(1 - a_t), \quad (9)$$

and the logit utility

$$z_t = -\alpha(\tilde{p}_t - \bar{p}) + \log(a_t + \epsilon). \quad (10)$$

The time-slot share, participation rate, and optional market expansion term are given by

$$s_t = \frac{e^{z_t}}{\sum_{\tau=1}^T e^{z_\tau}}, \quad (11)$$

$$m = \sigma \left[-\alpha \left(\sum_t s_t \tilde{p}_t - \bar{p} \right) \right], \quad (12)$$

$$g = 1 + \eta \max \left(\bar{p} - \sum_t s_t \tilde{p}_t, 0 \right). \quad (13)$$

Elastic demand is

$$F_t(\mathbf{p}) = \bar{F} g m s_t. \quad (14)$$

Total demand is $D_t = R_t + F_t$.

The baseline model does not feed QoS back into user utility. An ablation experiment additionally considers

$$z_t^{\text{feedback}} = z_t - \gamma[1 - q(u_t)], \quad (15)$$

solving for demand via fixed-point iteration. The experimental artifacts record convergence status and iteration count for each demand evaluation.

3.7 QoS Proxy, Profit, and In-Model Welfare

Let capacity be G and utilization $u_t = D_t/G$. The baseline QoS proxy is

$$q(u_t) = \begin{cases} 1, & u_t \leq \bar{u}, \\ \exp[-\kappa(u_t - \bar{u})^2], & u_t > \bar{u}. \end{cases} \quad (16)$$

This function is continuous. Section 5.9 constructs heuristic parameter mappings from single-GPU measurements, but joint calibration on production traces has not yet been completed.

In (16), $q(u_t)$ is interpreted in the profit function as the **effective fraction of completed service**: timeouts, truncations, or SLA violations reduce the number of billable requests. Accordingly, the revenue term uses $p_t D_t q(u_t)$. If a real platform charges for failed requests in full, the revenue model without the $q(u_t)$ multiplier should be used. Section 5 compares this alternative definition separately under a billing-mechanism ablation.

Platform profit is defined as

$$\begin{aligned} \Pi(\mathbf{p}) = & h \sum_t p_t D_t q(u_t) - h \sum_t w D_t \\ & - h \sum_t c_r (\bar{R}_t - R_t) - h \sum_t c_q [1 - q(u_t)] R_t, \end{aligned} \quad (17)$$

where h is the number of hours per period, w is the unit serving cost, and c_r and c_q are the rigid-user churn cost and QoS degradation cost, respectively. The in-model welfare metric is the sum of platform profit, rigid-user surplus, elastic-user log-sum surplus, and the external cost of carbon emissions. The experimental artifacts save each component separately.

4 Computational Algorithm

We present the QIDP (QoS-aware Intertemporal Dynamic Pricing) algorithm that solves the Stackelberg leader's problem (5). The algorithm combines gradient-based constrained optimization with an exact projection onto the feasible price set.

4.1 Bounded-Mean Projection

For a candidate price vector $\mathbf{x} \in \mathbb{R}^T$ that may violate the average-price constraint, we project it onto the feasible set $\mathcal{P}$ via a Lagrange-multiplier bisection.

Proposition 4 (Feasibility and Euclidean projection). *$\mathcal{P}$ is nonempty iff $\underline{p} \leq \bar{p} \leq \bar{p}$. For any $\mathbf{x} \in \mathbb{R}^T$, the projection $p_t(\lambda) = \min\{\bar{p}, \max\{\underline{p}, x_t + \lambda\}\}$ yields the unique minimizer of $\frac{1}{2}\|\mathbf{p} - \mathbf{x}\|_2^2$ over $\mathcal{P}$, where λ is found by bisection to satisfy $\frac{1}{T} \sum_t p_t(\lambda) = \bar{p}$.*

Proof. Nonemptiness: $(\bar{p}, \dots, \bar{p}) \in \mathcal{P}$. The projection is the KKT solution of a strictly convex quadratic program with a linear mean constraint and box bounds; the component-wise form follows from separable KKT conditions. The multiplier λ is unique unless multiple components lie at boundaries. $\square$

Algorithm 2 implements this procedure.

Table 2. BoundedMeanProjection

Input: $\mathbf{x} \in \mathbb{R}^T$, bounds $\underline{p}, \bar{p}$, target mean $\bar{p}$, tolerance $\varepsilon > 0$
Output: $\mathbf{p} \in \mathcal{P}$ satisfying $\frac{1}{T} \sum_t p_t = \bar{p}$
1. Set $\lambda_{\min} \leftarrow \underline{p} - \max_t x_t$, $\lambda_{\max} \leftarrow \bar{p} - \min_t x_t$ 2. while $\lambda_{\max} - \lambda_{\min} > \varepsilon$ do 3. $\lambda \leftarrow (\lambda_{\min} + \lambda_{\max})/2$ 4. $p_t \leftarrow \min\{\bar{p}, \max\{\underline{p}, x_t + \lambda\}\}$ for $t = 1, \dots, T$ 5. if $\frac{1}{T} \sum_t p_t < \bar{p}$ then $\lambda_{\min} \leftarrow \lambda$ else $\lambda_{\max} \leftarrow \lambda$ 6. end while 7. return $\mathbf{p}$ with $\lambda \leftarrow (\lambda_{\min} + \lambda_{\max})/2$

4.2 QIDP: QoS-Aware Intertemporal Dynamic Pricing

The main algorithm (Algorithm 3) performs multi-start SLSQP optimization of the platform's profit subject to the fixed-average-price constraint. To navigate the non-convex profit landscape, we initialize K candidate price vectors (one at the uniform baseline, the rest drawn randomly and projected onto $\mathcal{P}$), optimize each with SLSQP, and retain the best feasible candidate.

Table 3. QIDP: QoS-Aware Intertemporal Dynamic Pricing

Input: Periods T , capacity G , mean price $\bar{p}$, bounds $\underline{p}, \bar{p}$, QoS parameters $(\bar{u}, \kappa)$, profit parameters (h, w, c_r, c_q) , SLSQP tolerance τ , number of starts K
Output: Optimal price vector $\mathbf{p}^*$, associated profit Π^*
1. $\Pi^* \leftarrow -\infty$, $\mathbf{p}^* \leftarrow \text{None}$ 2. $\mathcal{S}_0 \leftarrow \{\bar{p} \cdot \mathbf{1}\} \cup \{\text{RandomUniform}(\underline{p}, \bar{p})^T\}_{k=1}^{K-1}$ 3. for each $\mathbf{p}^{(0)} \in \mathcal{S}_0$ do 4. $\mathbf{p} \leftarrow \text{BoundedMeanProjection}(\mathbf{p}^{(0)}, \underline{p}, \bar{p}, \bar{p})$ (Alg.2) 5. Solve $\max_{\mathbf{p} \in \mathcal{P}} \Pi(\mathbf{p})$ via SLSQP initialized at $\mathbf{p}$, with convergence tolerance τ 6. if SLSQP converged and $\mathbf{p}$ satisfies constraints then 7. $\mathbf{p} \leftarrow \text{BoundedMeanProjection}(\mathbf{p})$ (correct numerical drift) 8. if $\Pi(\mathbf{p}) > \Pi^*$ then $\Pi^* \leftarrow \Pi(\mathbf{p})$, $\mathbf{p}^* \leftarrow \mathbf{p}$ 9. end for 10. return $\mathbf{p}^*, \Pi^*$

Complexity. Each SLSQP iteration costs $O(T \cdot E)$, where E is the cost of evaluating the logit demand response $D(\mathbf{p})$ with QoS feedback (fixed-point iteration capped at $I_{\max}$ rounds). With K starts and $M_{\max}$ SLSQP iterations per start, total complexity is $O(K \cdot M_{\max} \cdot T \cdot I_{\max} \cdot (T + \text{cost of } q))$. In our implementation ($T \leq 64$, $K \leq 40$, $M_{\max} \leq 300$), runtimes range from 0.077 s to 12.3 s.

On global optimality. The profit function Π is non-convex in $\mathbf{p}$ due to the interaction of logit shares, QoS degradation, and rigid demand clamping. We therefore do not claim that Algorithm 3 returns the global optimum. For small instances ($T = 4$), we exhaustively enumerated 4495 grid points and confirmed that the SLSQP candidate is no worse than any grid point (and is within 0.26% of the best grid value). This provides a practical sanity check without constituting a global optimality certificate.

4.3 Comparison Policies

We evaluate three pricing policies within the Stackelberg framework:

1. **Uniform pricing (baseline):** $p_t = \bar{p}$ for all t . Serves as a no-scheduling benchmark.
2. **Myopic dynamic pricing:** Solves (5) with $\kappa = 0$ (ignoring QoS degradation), then evaluates the obtained price vector using the true κ . Isolates the benefit of “just dynamic pricing” from “QoS-aware pricing.”
3. **QoS-aware dynamic pricing (QIDP):** Algorithm 3 with the true κ . This is our proposed policy.

All policies are subject to the same fixed-average-price constraint to ensure fair comparison. The myopic policy uses the same multi-start SLSQP procedure as QIDP but with a flat QoS function during optimization.

5 Numerical Experiments

5.1 Experimental Setup and Reproducibility

The baseline configuration is $T = 8$, $h = 3$ hours, $G = 950$, $w = 0.40$, $\bar{p} = 0.80$, $\underline{p} = 0.45$, $\bar{p} = 2.10$, $\bar{u} = 0.82$, $\kappa = 15$, and $\alpha = 3.5$. All results are fully reproducible via automated experiment code. The mean-price constraint residuals do not exceed 9.99×10^{-13} across all 100 policy records, with zero solver failures.

5.2 RQ1: Can Constrained Dynamic Pricing Alleviate Congestion?

Table 4. Baseline comparison of pricing policies.

Policy	Profit	Welfare	Peak utilization	Min QoS
Uniform pricing	3600.20	14307.89	0.9855	0.6632
Myopic dynamic pricing	6176.45	15384.57	0.9105	0.8844
QoS-aware dynamic pricing	6498.68	15724.61	0.8434	0.9918

The price vector for QoS-aware dynamic pricing is

$$(0.45, 0.45, 0.45, 0.9212, 1.1852, 1.3992, 1.0944, 0.45).$$

Compared to uniform pricing, profit improves by 80.51% and welfare by 9.90%. Compared to myopic dynamic pricing, profit improves by 5.22%. Figure 1 shows the prices, utilization, and QoS.

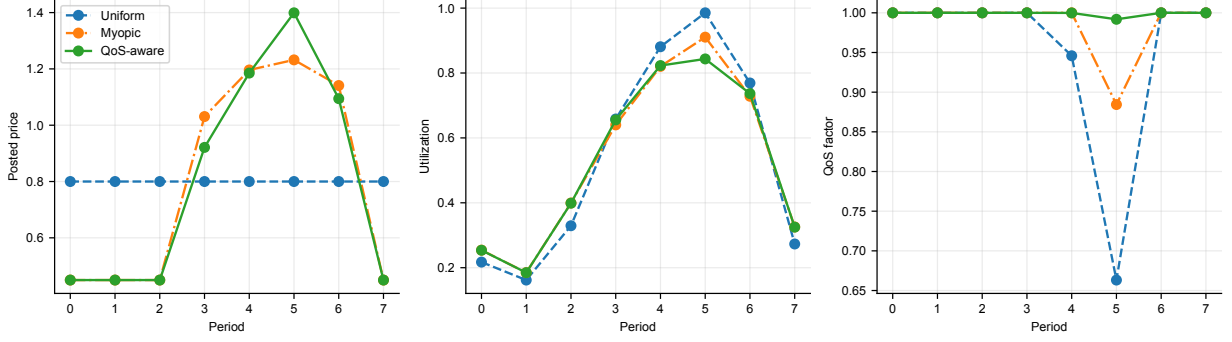


Figure 1. Baseline comparison across three pricing policies. Top: per-period prices; middle: utilization as fraction of capacity; bottom: QoS degradation factor. QoS-aware pricing shifts demand away from peak periods and maintains near-unity QoS throughout the horizon.

5.3 Solution Stability Check

The QoS-aware policy achieves the same profit across 10 random seeds, with a profit range of zero and a welfare range of 0.001241. This indicates that the current baseline search is stable under the employed multi-start configuration. It does not constitute a proof of global optimality.

5.4 Bill Protection and Coarse-Grid Reference

A fixed average price can still alter the demand-weighted bill. Table 5 adds the $\mathcal{P}_B(1)$ constraint and compares the posted bill with uniform pricing. Without bill protection, the posted bill increases by 19.53%; with protection, the posted bill does not exceed the uniform-pricing level, while profit still improves by 21.29%. Thus, the 80.51% profit gain in the main results should be interpreted as a scheduling benefit under a fixed average price, not as a bill-neutrality benefit.

Table 5. Effect of the fixed-average-price constraint with and without bill protection. The posted bill without protection exceeds the uniform baseline by 19.53%, while profit still improves by 21.29% even under strict bill protection ($\rho = 1$).

Policy	Profit	Posted bill	Bill ratio to uniform	Min QoS
Uniform pricing	3600.20	9748.05	1.0000	0.6632
Fixed average price	6498.68	11652.22	1.1953	0.9918
Fixed avg. price + bill protection	4366.70	9748.05	1.0000	0.9572

To provide an enumerable numerical reference, we reduce the number of periods to $T = 4$ and enumerate the 4495 grid points satisfying the mean constraint at a step size of 0.05. The best grid profit is 2918.94, while the multi-start SLSQP candidate profit is 2926.57 (0.261% higher). This check confirms that the candidate is at least no worse than the coarse grid.

5.5 RQ2: When Does QoS Modeling Yield Incremental Value?

Table 6 reveals a boundary that must be stated explicitly: even when $\kappa = 0$, dynamic pricing yields a 37.75% gain over uniform pricing. This portion arises from heterogeneity in time-slot preferences, price sensitivity, and rigid-user willingness to pay, not from QoS degradation. The incremental value of QoS modeling should be measured by the difference between QoS-aware and myopic dynamic pricing.

Table 6. Sensitivity to QoS degradation severity κ . The incremental value of explicit QoS modeling is measured as the profit gain of QoS-aware pricing over myopic pricing (parentheses). This gain increases from 0% at $\kappa = 0$ to 12.53% at $\kappa = 30$.

κ	Profit gain vs. uniform	Profit gain vs. myopic	Min QoS
0	37.75%	0.00%	1.0000
5	49.96%	0.99%	0.9871
15	80.51%	5.22%	0.9918
30	136.23%	12.53%	0.9952

Table 7. Sensitivity of the QoS-aware profit increment to key assumptions. The advantage over myopic dynamic pricing is largest under tight capacity ($G = 750$, +293.78%) and disappears under abundant capacity ($G = 1200$, 0%).

Experiment	Parameter value	Profit increment
Fixed average price	0.60/0.80/1.00/1.20	9.90/5.22/2.46/2.74%
Capacity	750/950/1200	293.78/5.22/0.00%
Market expansion coefficient	0/2.5/6.0	4.88/5.22/5.71%
QoS feedback weight	0/1/2	5.22/4.99/4.80%

5.6 RQ3: Are Conclusions Robust to Key Assumptions?

When $G = 750$, uniform pricing yields a negative profit, so percentage gains over uniform pricing would be misleading. Table 7 instead compares QoS-aware pricing with myopic dynamic pricing. When $G = 1200$, the two policies achieve nearly identical profit, indicating that the incremental value of explicit QoS modeling vanishes when capacity is abundant.

5.7 Rule Baselines, Billing Ablation, and Scalability

To verify that the QoS-aware multi-start search outperforms more than just a weak baseline, we add peak-valley rules and queue-aware rules. Both rule types adjust per-period prices based on the rigid load profile, then project back onto the mean constraint via (4). Table 8 reports the results.

Table 8. Rule-based pricing baselines and billing mechanism ablation. The QoS-aware multi-start search outperforms both heuristic rules, and the choice of billing rule (full billing vs. SLA penalty) affects the optimal load distribution.

Policy or billing mechanism	Profit	Welfare	Min QoS
Uniform pricing	3600.20	14307.89	0.6632
Peak-valley rule	5633.67	15331.25	0.8533
Queue-aware rule	6489.51	15697.14	0.9897
QoS-aware multi-start search	6498.68	15724.61	0.9918
Full billing	6575.24	15815.07	0.9423
SLA penalty billing	6545.08	15786.57	0.9650

5.8 Single-GPU System Measurements

Supplementary single-GPU load experiments use Ollama 0.24.0 on RTX 4090 with a 9B-parameter Q4_K_M quantized GGUF model.

Table 9 shows that total throughput increases by 8.55% when concurrency rises from 1 to 2, then largely saturates; mean TTFT and P95 TTFT increase steadily with concurrency. The primary

Table 9. Single-GPU Ollama load experiment on RTX 4090 with a 9B-parameter quantized model. Throughput saturates beyond two concurrent requests while TTFT increases steadily, indicating queueing-dominated congestion.

Concurrency	Completion rate	Mean TTFT (s)	P95 TTFT (s)	TPOT (ms)	Throughput (token/s)
1	100%	0.155	0.167	9.456	91.46
2	100%	1.232	1.421	9.413	99.28
4	100%	3.296	4.032	9.426	98.75
8	100%	6.136	9.175	9.464	98.63

congestion manifestation in this backend is queueing growth rather than TPOT degradation.

5.9 vLLM Primary Backend Overload Experiments

Repeated-overload scans using vLLM 0.22.0 with Qwen2.5-0.5B-Instruct on RTX 4090.

Table 10. vLLM repeated-overload scan on RTX 4090 with Qwen2.5-0.5B. The 0.5 s SLA attainment rate drops from 100.0% at 64 concurrency to 50.0% at 512 concurrency. Parentheses show 95% confidence half-widths over 5 repetitions of 512 requests each.

Concurrency	Mean TTFT (s)	P95 TTFT (s)	Throughput (token/s)	0.5 s SLA rate
64	0.049 (0.001)	0.073 (0.003)	16125.98 (246.35)	100.0 (0.0)%
128	0.084 (0.004)	0.139 (0.006)	19854.91 (568.53)	100.0 (0.0)%
224	0.122 (0.005)	0.225 (0.029)	20128.54 (736.48)	100.0 (0.0)%
384	0.544 (0.023)	1.617 (0.077)	18983.53 (453.70)	63.5 (0.7)%
512	0.632 (0.038)	1.588 (0.063)	19266.09 (291.99)	50.0 (0.0)%

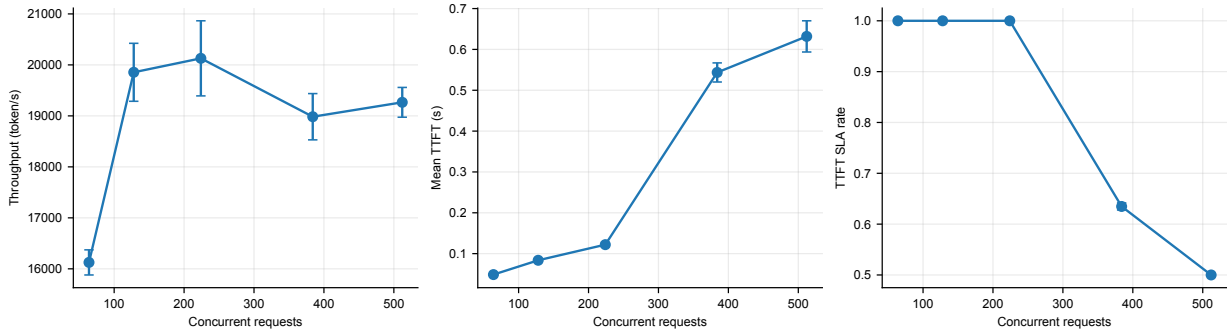


Figure 2. vLLM overload curves for Qwen2.5-0.5B on RTX 4090. Left: mean and P95 TTFT versus concurrency; right: throughput and 0.5 s SLA attainment rate. The SLA rate falls sharply between 224 and 384 concurrent requests, confirming a measurable capacity boundary. Error bars show 95% confidence half-widths.

5.10 Empirical QoS Heuristic Mapping and Robustness Check

Using TTFT SLA attainment rate as the empirical QoS observation, we normalize the last concurrency level with an SLA rate of at least 99% as $u = 1$, then construct heuristic scenario mappings via (16).

Substituting both sets of heuristic parameters into the economic model and re-optimizing prices yields a profit of 6649.24 and a minimum QoS of 1.0000 in both cases, compared to 6498.68 for the baseline synthetic proxy.

Table 11. Empirical QoS heuristic mapping from vLLM repeated-overload scans. The normalized healthy capacity is 224 concurrent requests for both models. The 0.5B model shows higher fitting error (RMSE 0.0683) due to the 384-concurrency transition point.

Model	Normalized threshold	Degradation severity	RMSE
Qwen2.5-0.5B-Instruct	1.000	0.517	0.0683
Qwen2.5-3B-Instruct	1.058	0.942	6.33×10^{-10}

6 Discussion

The current version primarily uses algorithmic simulation, supplemented by single-GPU backend measurements using vLLM and Ollama to validate the congestion phenomenon. These measurements cannot substitute for system experiments across a broader range of model sizes, multiple backends, and multi-GPU configurations. The electricity dynamic pricing literature provides a methodological reference, but inference serving does not involve grid power flow, nodal balance, or wholesale market clearing constraints.

Key limitations include: (1) the QoS function remains a parametric proxy; (2) the welfare metric depends on synthetic parameters; (3) multi-start SLSQP does not provide a global optimality certificate for the non-convex problem; (4) the aggregate logit response is not a proof of pure-strategy Nash equilibrium; (5) multi-platform competition is excluded; (6) the demand curve is derived from synthetic parameters; (7) a fixed average price is not equivalent to bill neutrality; (8) price-sensitivity analysis does not demonstrate distributional fairness.

7 Conclusion

We adapt the intertemporal scarcity-pricing mechanism and leader–follower hierarchical structure from electricity demand response to token services. The platform selects a price vector under a fixed-average-price constraint, and elastic users migrate demand in response to prices and time-slot preferences. On the theoretical side, we prove the existence of a nonempty feasible price region and an optimal platform solution; on the algorithmic side, we implement an exact bounded-mean projection, multi-start optimization, and traceable experimental artifacts. Synthetic experiments answer three research questions: constrained dynamic pricing can shift demand and alleviate congestion; the profit advantage of explicit QoS modeling over myopic pricing increases with degradation severity and disappears when capacity is abundant; and this incremental value maintains interpretable boundaries across multiple sensitivity experiments. Under the additional bill-protection constraint, profit still improves by 21.29% over uniform pricing. These conclusions are confined to synthetic demand and a parametric QoS proxy.

8 Code and Artifact Availability

All code, experimental configurations, and analysis scripts required to reproduce the results are included in the supplementary materials. The official public repository URL will be made available upon publication.

A Baseline Parameters

Table 12. Synthetic economic model baseline parameters.

Parameter	Baseline value	Description
T, h	8, 3	Number of periods and hours per period
G, w	950, 0.40	Capacity and unit serving cost
$\bar{p}, \underline{p}, \bar{p}$	0.80, 0.45, 2.10	Mean price and price bounds
$\bar{u}, \kappa$	0.82, 15	QoS threshold and degradation severity
α, β	3.5, 5.0	Elastic and rigid price sensitivity
$v_r, \bar{F}$	1.80, 400	Rigid willingness-to-pay ceiling and elastic market size
c_s, η	0.20, 2.50	Switching cost and market expansion coefficient
c_r, c_q	0.60, 0.35	Rigid churn cost and QoS degradation cost

The rigid load vector and time-slot preference vector are

$$(200, 150, 300, 600, 800, 900, 700, 250), \quad (0.3, 0.2, 0.5, 0.8, 1.0, 1.0, 0.9, 0.4),$$

respectively. The carbon emission rate and unit carbon cost are 0.20 and 0.05. The bill-protection scenario uses $\rho = 1$.

B Complete Sensitivity Range

Table 13. Phase one complete sensitivity range.

Experiment	Parameter range	Primary purpose
Fixed average price	0.60, 0.80, 1.00, 1.20	Examine price constraint boundary
QoS degradation severity	0, 5, 15, 30	Separate dynamic pricing gains from QoS incremental value
QoS threshold	0.75, 0.82, 0.90	Examine congestion trigger location
Capacity	750, 950, 1200	Examine tight vs. abundant capacity
Price sensitivity	1.0, 3.5, 8.0	Examine elastic user response
Switching cost	0.05, 0.20, 0.80	Examine intertemporal migration resistance
Market expansion coefficient	0, 2.5, 6.0	Rule out market expansion as the sole source
Churn cost	0, 0.60, 1.20	Examine long-term cost assumptions
QoS feedback weight	0, 1, 2	Examine demand feedback variant

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